

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2018

PART II

TRANSFER PROCESSES 2

SOLUTIONS

SECTION I

Question 1

(a)

$$\begin{aligned}\text{Re}_\delta|_{x=L} &= \frac{4\Gamma|_{x=L}}{\mu_l} \\ \text{Re}_\delta|_{x=L} &= \frac{4\dot{m}}{(\pi D_o)\mu_l} \\ \dot{m} &= \frac{Q}{h_{lv}} = \frac{\bar{\alpha}(\pi D_o L)(T_w - T_{sat})}{h_{lv}} \\ \therefore \text{Re}_\delta|_{x=L} &= \frac{4}{(\pi D_o)\mu_l} \frac{\bar{\alpha}(\pi D_o L)(T_w - T_{sat})}{h_{lv}} = \frac{4\bar{\alpha}(L)(T_w - T_{sat})}{\mu_l h_{lv}} \\ \bar{\alpha} &= \frac{\mu_l h_{lv} \text{Re}_\delta|_{x=L}}{4(L)(T_w - T_{sat})} = \frac{2.79 \times 10^{-4} (2257) \times 10^3 (80)}{4(0.5)(8)} = 3148.5 \text{ W m}^{-2} \text{ K}^{-1}\end{aligned}$$

[10 marks]

(b)

$$\begin{aligned}\bar{\alpha}_H &= \frac{0.729}{0.943} \left(\frac{L}{D_o} \right)^{1/4} \quad \bar{\alpha}_v = 0.773 \left(\frac{0.5}{0.06} \right)^{1/4} \quad \bar{\alpha}_v = 1.313 \bar{\alpha}_v \\ &= 4134 \text{ W m}^{-2} \text{ K}^{-1}\end{aligned}$$

The average heat transfer coefficient would be increased by 31.3% compared to the vertical tube.

[4 marks]

(c) For shear-controlled condensation, local heat transfer coefficient is given by (from equation sheet),

$$\begin{aligned}\alpha &= \sqrt{\frac{k_l^2 \tau_i \rho_l}{2\Gamma \mu_l}} \\ \Gamma &= \rho_l \bar{u} \delta; \alpha = \frac{k_l}{\delta} \\ \therefore \frac{k_l}{\delta} &= \sqrt{\frac{k_l^2 \tau_i \rho_l}{2(\rho_l \bar{u} \delta) \mu_l}} \\ \frac{1}{\delta} &= \sqrt{\frac{\tau_i}{2\bar{u} \delta \mu_l}} \\ \delta &= \frac{2\bar{u} \mu_l}{\tau_i} = \frac{2(0.6)(2.79 \times 10^{-4})}{4.0} = 8.37 \times 10^{-5} \text{ m} \\ \alpha &= \frac{k_l}{\delta} = \frac{0.68}{8.37 \times 10^{-5}} = 8124.3 \text{ W m}^{-2} \text{ K}^{-1}\end{aligned}$$

[11 marks]

Question 2

- (a) From the psychrometric chart, the relative humidity is 20%, the wet-bulb temperature is 10°C and the specific enthalpy is 29.5 kJ (kg dry air)⁻¹.

[3 marks]

(b)

The initial air stream is at condition 1, the humidified air stream is at condition 2. Known conditions are:

$$T_1=21^\circ\text{C}; \omega_1=0.003 \text{ kg (kg dry-air)}^{-1}; h_1=29.5 \text{ kJ (kg dry air)}^{-1}; \dot{m}_{a1} = 2 \text{ kgs}^{-1}$$
$$h_w=2676 \text{ kJ kg}^{-1}; \text{RH}_2=50\%$$

- (i) From mass and energy balances, we have

$$\begin{aligned}\dot{m}_{a1} &= \dot{m}_{a2} \\ \dot{m}_{a1}\omega_1 + \dot{m}_w &= \dot{m}_{a2}\omega_2 \\ \therefore \dot{m}_w &= \dot{m}_{a1}(\omega_2 - \omega_1) \\ \dot{m}_{a1}h_1 + \dot{m}_wh_w &= \dot{m}_{a2}h_2 \\ \dot{m}_wh_w &= \dot{m}_{a1}(h_2 - h_1) \\ \therefore \frac{h_2 - h_1}{\omega_2 - \omega_1} &= h_w \\ \frac{h_2 - 29.5}{\omega_2 - 0.003} &= 2676\end{aligned}$$

Choose any point that satisfy the above relation, e.g. $\omega_2=0.007$, then $h_2=40.2$. Find point 2' on the chart and draw a straight line between points 1 and 2'. The intersecting point between the straight line and the RH=50% line is the final point 2. Reading from the chart we have $\omega_2=0.009 \text{ kg (kg dry-air)}^{-1}$

$$\dot{m}_w = \dot{m}_{a1}(\omega_2 - \omega_1) = 2(0.009 - 0.003) = 0.012 \text{ kgs}^{-1}$$

[10 marks]

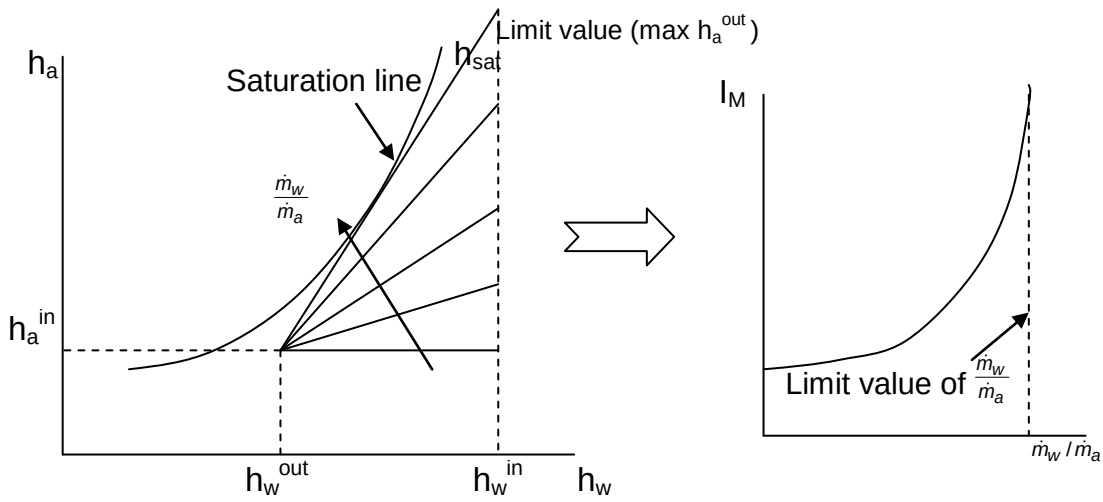
- (ii) From the chart we can also read the temperature at point, $T_2=23.5^\circ\text{C}$

[2 marks]

- (c) From energy balance:

$$\begin{aligned}\dot{m}_a(h_a^{out} - h_a^{in}) &= \dot{m}_w(h_w^{in} - h_w^{out}) \\ \therefore h_a^{out} &= h_a^{in} + \frac{\dot{m}_w}{\dot{m}_a}(h_w^{in} - h_w^{out})\end{aligned}$$

So that h_a^{out} increases with an increase in $\dot{m}_w/\dot{m}_a$. By constructing operating lines, it can be found that $h_{\text{sat}} - h_a$ decreases when $\dot{m}_w/\dot{m}_a$ is increased. Hence I_M will increase with an increase in $\dot{m}_w/\dot{m}_a$. This can be illustrated by the figure given below.



[10 marks]

SECTION II

Question 3

(a) The mean beam length represents an average thickness of gas encountered by all beams of radiation between two surfaces. For a cube of side length 3 m, we have: $L_m = \frac{3.6(3 \times 3 \times 3)}{6(3 \times 3)} = 1.8m$

[4 marks]

(b)

There are two radiation surfaces:

Sphere: $\varepsilon_1=0.8$, $\rho_1=1-0.8=0.2$, $d_1=0.1$ m, $T_1=900$ K

Enclosure: $\varepsilon_2=0.6$, $\rho_2=1-0.6=0.4$

(i) The overall heat balance for the sphere is:

$$\begin{aligned} Q_{in} + A_1 H_1 &= Q_{out} + A_1 W_1 \\ \therefore A_1 (W_1 - H_1) &= Q_{in} \\ W_1 - H_1 &= \frac{Q_{in}}{A_1} = \frac{800}{\pi(0.1)^2} = 25478 W m^{-2} \end{aligned}$$

[5 marks]

(ii) Performing radiation balances on the sphere and enclosure we have

$$\begin{aligned} W_1 &= \varepsilon_1 E_{b1} + \rho_1 [W_1 F_{11} + W_2 F_{12}] \\ W_2 &= \varepsilon_2 E_{b2} + \rho_2 [W_1 F_{21} + W_2 F_{22}] \end{aligned}$$

For the sphere $F_{11}=0$, so that $F_{12}=1 - F_{11}=1$.

Also, from reciprocity and summation, $F_{21}=F_{12}(A_1/A_2)=1(\pi 0.1^2/0.4)=0.0785$, $F_{22}=1 - F_{21}=0.9215$

Now

$$\begin{aligned} W_1 &= 0.8 E_{b1} + 0.2 W_2 \\ W_2 &= 0.6 E_{b2} + 0.4(0.0785 W_1 + 0.9215 W_2) \end{aligned}$$

Solutions are

$$\begin{aligned} W_1 &= 0.8080 E_{b1} + 0.1920 E_{b2} \\ W_2 &= 0.0402 E_{b1} + 0.9598 E_{b2} \end{aligned}$$

We also have

$$H_1 = W_1 F_{11} + W_2 F_{12} = W_2$$

Substituting into the overall heat balance equation,

$$W_1 - H_1 = W_1 - W_2 = 25478 \text{ Wm}^{-2}$$

or

$$(0.8080 - 0.0402)E_{b1} + (0.1920 - 0.9598)E_{b2} = 25478$$

$$0.7678E_{b1} - 0.7678E_{b2} = 25478$$

$$T_1^4 - T_2^4 = \frac{25478}{0.7678(5.67 \times 10^{-8})} = 5852.4 \times 10^8$$

$$T_2^4 = (900)^4 - 5852.4 \times 10^8 = 708.6 \times 10^8$$

$$T_2 = 516 \text{ K}$$

[16 marks]

Question 4

(a)

The view factor from a surface i to a surface j is defined as the fraction of the radiation leaving surface i that is directly intercepted by surface j .

The view factors between the sphere and an infinitely large plane are:

$$F_{ss}=F_{pp}=0, F_{sp}=1/2, F_{ps}=1/2(A_s/A_p) \approx 0 \text{ (Since } A_p \gg A_s)$$

[6 marks]

(b)

(i) There are two surfaces, the upper plate 1 and the lower plate 2. $\epsilon_1=\epsilon_2=0.6$, $\rho_1=\rho_2=1-0.6=0.4$
Performing radiation balances on the two plates we have

$$W_1 = \epsilon_1 E_{b1} + \rho_1 (W_1 F_{11} + W_2 F_{12})$$

$$W_2 = \epsilon_2 E_{b2} + \rho_2 (W_1 F_{21} + W_2 F_{22})$$

$$\text{Since } F_{11}=F_{22}=0, \text{ and } F_{12} = F_{21} = \frac{2 \times \sqrt{1^2 + 2^2} - 2 \times 1}{2 \times 2} = \frac{\sqrt{5} - 1}{2} = 0.618$$

$$W_1 = \epsilon_1 E_{b1} + \rho_1 W_2 F_{12} = 0.6 E_{b1} + 0.2472 W_2$$

$$W_2 = \epsilon_2 E_{b2} + \rho_2 W_1 F_{12} = 0.6 E_{b2} + 0.2472 W_1$$

Solutions are

$$W_1 = 0.6390 E_{b1} + 0.1580 E_{b2}$$

$$W_2 = 0.1580 E_{b1} + 0.6390 E_{b2}$$

Net radiation from the upper plate is

$$Q_{net,1} = A_1 F_{12} (W_1 - W_2)$$

$$q_{net,1} = \frac{Q_{net,1}}{A_1} = (0.618)(0.6390 E_{b1} + 0.1580 E_{b2} - 0.1580 E_{b1} - 0.6390 E_{b2})$$

$$= (0.618)(0.481)(E_{b1} - E_{b2}) = 0.297(E_{b1} - E_{b2}) = 0.297(5.67 \times 10^{-8})(T_1^4 - T_2^4)$$

[12 marks]

Performing overall heat balance on the lower plate:

$$H_2 = W_2$$

$$H_2 = W_1 F_{21}$$

$$\therefore 0.618 W_1 = W_2$$

$$0.618(0.6390E_{b1} + 0.1580E_{b2}) = 0.1580E_{b1} + 0.6390E_{b2}$$

$$0.2369E_{b1} = 0.5414E_{b2}$$

$$T_2^4 = \frac{0.2369}{0.5414} T_1^4 = 0.4376(1000)^4$$

$$T_2 = 813.3K$$

[7 marks]

2019
TRANSFER PROCESSES 2
MODEL SOLUTIONS

QUESTION 1

- (a) Edge effects are negligible and therefore $w \neq f(y)$ (the profile is uniform in the y-direction). Because the flow is unidirectional ($u = v = 0$), we obtain from the continuity equation that $\frac{\partial w}{\partial z} = 0$, i.e. $w \neq f(z)$. Therefore, $w = w(x)$.

Because $u = 0$, the x-component momentum equation leads to: $dp/dx = 0$, i.e. $p = p(y, z)$.

Because $v = 0$, the y-component momentum equation leads to: $dp/dy = 0$, i.e. $p = p(z)$

[4 marks]

- (b) z-component momentum equation:

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left[\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right] + g_z$$

- Steady-state: $\partial w / \partial t = 0$ [drop 1st term LHS]
- Unidirectional flow: $u = v = 0$ [drop 2nd and 3rd terms LHS]
- From the continuity equation (above): $\partial w / \partial z = 0$ [drop 4th LHS]
- From the continuity equation (above): $w = w(x)$ [drop 3rd and 4th terms RHS]

The simplified equation reads as follows:

$$\mu \frac{\partial^2 w}{\partial x^2} = \frac{\partial p}{\partial z} - \rho g_z$$

[5 marks]

- (c) Because $\frac{\partial p}{\partial z} - \rho g_z$ is constant, both the LHS and the RHS of the above equation are constant. Moreover, ordinary derivatives can be used because $w = w(x)$ and $p = p(z)$. The equation to be integrated reads as follows:

$$\frac{d^2 w}{dx^2} = \frac{1}{\mu} \left(\frac{dp}{dz} - \rho g_z \right)$$

Upon integration the following expression is obtained:

First integration:

$$\frac{dw}{dx} = \frac{1}{\mu} \left(\frac{dp}{dz} - \rho g_z \right) x + A$$

Second integration:

$$w = \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) x^2 + Ax + B$$

The following two BCs apply:

At $x = +D/2$: $w = 0$ and at $x = -D/2$: $w = W_0$, leading to the following two expressions:

$$0 = \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) \frac{D^2}{4} + A \frac{D}{2} + B \quad \text{and} \quad W_0 = \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) \frac{D^2}{4} - A \frac{D}{2} + B$$

The constant A is obtained from their subtraction: $A = -W_0/D$

The constant B is then obtained from the first expression: $B = \frac{W_0}{2} - \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) \frac{D^2}{4}$

Combining both expressions in $w = \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) x^2 + Ax + B$ and rearranging, gives the following resulting expression:

$$w = \frac{W_0}{2} \left(1 - \frac{x}{d} \right) - \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) d^2 \left[1 - \left(\frac{x}{d} \right)^2 \right]$$

where $d = D/2$ has been used.

[5 marks]

(d) The average velocity through the slit is can be obtained from the following expression:

$$\bar{w} = \frac{\int_{-d}^d \int_0^L w(x) dx dy}{\int_{-d}^d \int_0^L dx dy} = \frac{\int_{-d}^d w(x) dx}{2d}$$

For convenience we introduce $x_D = x/d$ (accordingly, $dx_D = dx/d$). It follows that:

$$\bar{w} = \frac{1}{2} \int_{-1}^1 w(x_D) dx_D = \frac{1}{2} \left\{ \frac{W_0}{2} \int_{-1}^1 (1 - x_D) dx_D - \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) d^2 \int_{-1}^1 (1 - x_D^2) dx_D \right\}$$

After integrating, we obtain the following expression:

$$\bar{w} = \frac{1}{2} \int_{-1}^1 w(x_D) dx_D = \frac{1}{2} \left\{ W_0 - \frac{2}{3\mu} \left(\frac{dp}{dz} - \rho g_z \right) d^2 \right\} = \frac{W_0}{2} - \frac{\left(\frac{dp}{dz} - \rho g_z \right) d^2}{3\mu}$$

Note that the sign of $\bar{w}$ will depend on the relative strength of the two terms in the equation (plate velocity vs. pressure gradient).

[2 marks]

(e) The position of the maximum velocity is found by solving $dw/dx = 0$, i.e.

$$0 = \frac{1}{\mu} \left(\frac{dp}{dz} - \rho g_z \right) x - \frac{W_0}{2d}$$

$$x_{max} = \frac{\mu W_0}{2d \left(\frac{dp}{dz} - \rho g_z \right)}$$

For $W_0 > 0$, the maximum velocity is therefore attained in the positive midplane, as opposed to the centre of the slit for ($W_0 = 0$).

[2 marks]

- (f) The stress profile is obtained from the following expression: $\tau_{zx} = \mu \frac{dw}{dx}$, i.e.

$$\tau_{zx} = \left(\frac{dp}{dz} - \rho g_z \right) x - \frac{\mu W_0}{2d}$$

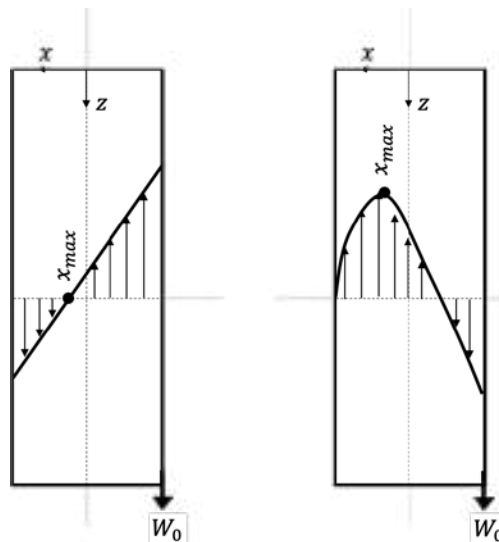
which further implies that the stress varies linearly with distance. We note the following points:

- at x_{max} : $\tau_{zx} = 0$
- at $x = 0$: $\tau_{zx} = -\frac{\mu W_0}{2d}$
- at $x = d$: $\tau_{zx} = \left(\frac{dp}{dz} - \rho g_z \right) d - \frac{\mu W_0}{2d}$ (we assume here that it is positive)
- at $x = -d$: $\tau_{zx} = -\left(\frac{dp}{dz} - \rho g_z \right) d - \frac{\mu W_0}{2d}$

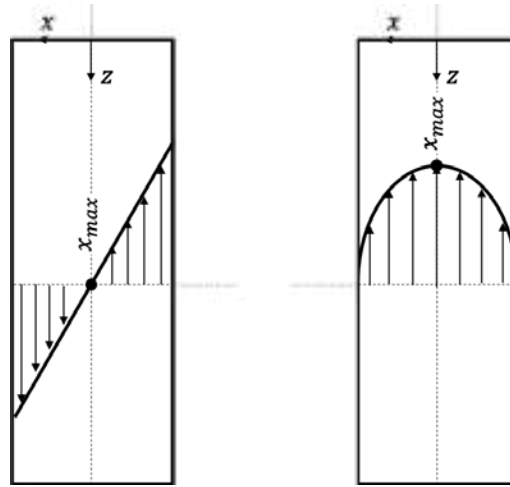
For the velocity profile, the following points are obtained:

- at $x = 0$: $w = \frac{W_0}{2} - \frac{1}{2\mu} \left(\frac{dp}{dz} - \rho g_z \right) d^2$ (we assume here that it is negative)
- at $x = d$: $w = 0$
- at $x = -d$: $w = W_0$

The following sketch is obtained (note the reversal of flow direction):



For the situation of $W_0 = 0$, the following sketch is obtained:

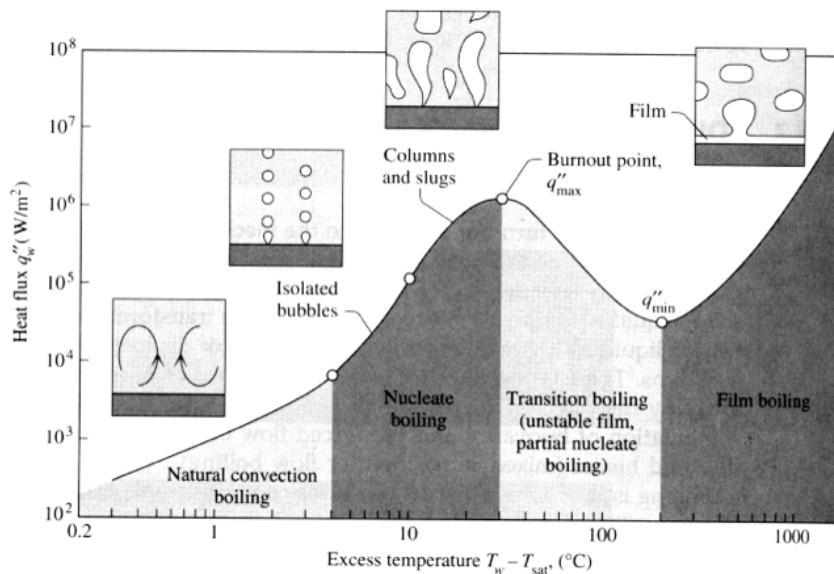


[7 marks]

QUESTION 2

(a)

(i) Sketch of boiling curve:



(ii) In natural convection boiling, the system remains in a single phase and heat is transferred by convection in the liquid. In nucleate boiling, heat transfer is promoted by the rise of bubbles, which causes forced convection in the liquid. In film boiling, there is convective heat transfer through a continuous vapour film; for sufficiently high temperature ($T_w > 300$ C) there is also a contribution from radiation across the vapour film.

(iii) The burnout point should be avoided in the design of boilers because near this point a small increase in the heat flux (in power-controlled mode) can lead to a very large increase in temperature of the heater surface. The temperature can exceed the melting temperature of

the metal, causing damage to equipment. This can be prevented if the equipment is operated in temperature-controlled mode.

[5 marks]

(b)

- (i) Excess temperature is $\Delta T_E = T_w - T_{sat} = 109^\circ\text{C} - 100^\circ\text{C} = 9^\circ\text{C}$. Boiling regime is nucleate boiling, hence Rohsenow correlation applies:

$$q = \mu_l h_{lv} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{\frac{1}{2}} \left[\frac{c_{p,l} \Delta T_E}{C_{sf} h_{lv} Pr^s} \right]^3.$$

Because the water in the reboiler is saturated, we use properties of water at $T_{sat} = 100^\circ\text{C}$. It is incorrect to take the properties of water at 109°C , which is the temperature of the wall, not of the water. It is also incorrect to take the properties of water at $T_{mean} = \frac{100^\circ\text{C} + 109^\circ\text{C}}{2}$! From the table, at $T_{sat} = 100^\circ\text{C}$ we have:

$$\begin{aligned}\mu_l &= 0.279 \times 10^{-3} \text{ Pa s} \\ h_{lv} &= 2.257 \times 10^6 \text{ J/kg} \\ \rho_l &= 957.9 \text{ kg m}^{-3} \\ \rho_v &= 0.5956 \text{ kg m}^{-3} \\ \sigma &= 58.9 \times 10^{-3} \text{ N/m} \\ c_{p,l} &= 4217 \text{ J kg}^{-1} \text{ K}^{-1} \\ k_l &= 0.68 \text{ W m}^{-1} \text{ K}^{-1}\end{aligned}$$

First we calculate the Prandtl number $Pr = \frac{\mu_l c_{p,l}}{k_l} = \frac{0.279 \times 10^{-3} \times 4217}{0.68} = 1.73$

and then plug into:

$$\begin{aligned}q &= (0.279 \times 10^{-3})(2.257 \times 10^6) \left[\frac{9.81(957.9 - 0.5956)}{58.9 \times 10^{-3}} \right]^{1/2} \times \\ &\times \left[\frac{4217 \times 9}{0.0133 (2.257 \times 10^6) 1.73^1} \right]^3 \text{ W m}^{-2} = 98 \text{ kW m}^{-2}\end{aligned}$$

Multiply by total area of heater, which is equivalent to a cylinder with length $L = 30 \text{ m}$ and diameter $D = 5 \text{ cm}$:

$$Q = q \times (\pi DL) = (98 \times 10^3)(\pi \times 0.05 \times 30) \text{ W} = 462 \text{ kW}$$

- (ii) At steady state, all the heat transferred to the liquid results in water evaporation. Rate of evaporation $\dot{m}_v$ obeys:

$$Q = \dot{m}_v h_{lv} ;$$

$$\dot{m}_v = \frac{Q}{h_{lv}} = \frac{462 \times 10^3}{2.257 \times 10^6} \text{ kg s}^{-1} = 0.205 \text{ kg s}^{-1}.$$

(c)

Excess temperature is $\Delta T_E = T_w - T_{sat} = 350^\circ\text{C} - 100^\circ\text{C} = 250^\circ\text{C}$. Boiling regime is film boiling. Since $T_w > 300^\circ\text{C}$, there are two contributions to heat transfer, one from radiative heat transfer through vapour film, and one from convection in vapour film:

$$q_{total} = \alpha_{total} \Delta T_E,$$

$$\text{with } \begin{cases} \alpha_{total} = \alpha_{conv} + \frac{3}{4} \alpha_{rad} & \text{if } \alpha_{rad} < \alpha_{conv} \\ \alpha_{total}^{4/3} = \alpha_{conv}^{4/3} + \alpha_{rad} \alpha_{total}^{1/3} & \text{if } \alpha_{rad} > \alpha_{conv} \end{cases}$$

Calculate convective heat transfer coefficient. Use the mean temperature of the vapour film

$$T_{film} = \frac{350^\circ\text{C} + 100^\circ\text{C}}{2} = 225^\circ\text{C} \text{ to estimate the properties of the vapour. The values given in}$$

Property Table are for $T_v = 220^\circ\text{C}$ which is a reasonable approximation (no need to interpolate, but it is not incorrect to interpolate). Properties of liquid are taken at $T_{sat} = 100^\circ\text{C}$.

The corrected latent heat of vaporisation (correction needed when $\Delta T_E > 10^\circ\text{C}$) is:

$$h'_{lv} = h_{lv} + 0.8 c_{p,v} \Delta T_E = (1.87 \times 10^6 + 0.8 \times 1983 \times 250) \text{ J/kg} = 2.27 \times 10^6 \text{ J/kg}.$$

To determine $\alpha_{conv} = \frac{Nu_D k_v}{D}$, compute Nusselt number:

$$\begin{aligned} Nu_D &= C \left[\frac{(\rho_l - \rho_v) g h'_{lv} D^3}{\nu_v k_v \Delta T_E} \right]^{\frac{1}{4}} = \\ &= 0.62 \left[\frac{(957.9 - 10.84) \times 9.81 \times (2.27 \times 10^6) (0.05)^3}{0.398 \times 10^{-4} \times 0.0358 \times 250} \right]^{\frac{1}{4}} = 182 \end{aligned}$$

where $C = 0.62$ for horizontal cylinders.

$$\text{Heat transfer coefficient is } \alpha_{conv} = \frac{182 \times 0.0358}{0.05} \text{ W m}^{-2} \text{ K}^{-1} = 130.20 \text{ W m}^{-2} \text{ K}^{-1}.$$

Radiative heat transfer coefficient:

$$\begin{aligned} \alpha_{rad} &= \frac{\varepsilon \sigma (T_w^4 - T_{sat}^4)}{T_w - T_{sat}} = \frac{0.75 \times 5.67 \times 10^{-8} (623.15^4 - 373.15^4)}{250} \text{ W m}^{-2} \text{ K}^{-1} = \\ &= 22.35 \text{ W m}^{-2} \text{ K}^{-1}. \end{aligned}$$

Hence

$$\alpha_{total} = \alpha_{conv} + \frac{3}{4} \alpha_{rad} = 147 \text{ W m}^{-2} \text{ K}^{-1}$$

$$q_{total} = \alpha_{total} \Delta T_E = 147 \times 250 \text{ W m}^{-2} = 36.75 \text{ kW m}^{-2}$$

Total heat transfer rate

$$Q = q_{total} \times (\pi DL) = 36.75 \times 10^3 (\pi \times 0.05 \times 30) \text{ W} = 173 \text{ kW}$$

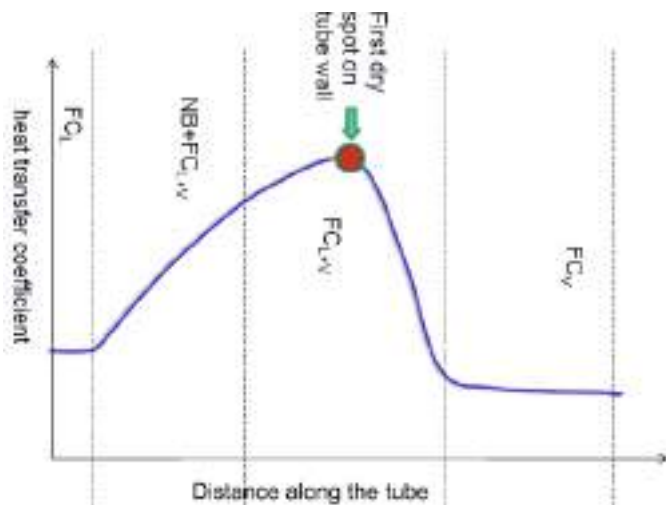
[7 marks]

(d)

Answers to (i) / (ii) from bottom to top of pipe:

- single phase liquid / forced convection in liquid
- bubbly flow / subcooled and then saturated nucleate boiling
- slug flow and annular flow / saturated nucleate boiling
- annular flow / forced convection through liquid film; heat transfer coefficient reaches maximum and then decreases when first dry spot appears on the wall
- drop flow / liquid-deficient region
- single phase vapour / forced convection in vapour

(iii)



[5 marks]

QUESTION 3

(a)

The N pipes are treated as one object (p), and the furnace wall is denoted by f . There are four view factors: F_{pp} , F_{pf} , F_{fp} and F_{ff} . The total radiative flux leaving an object is given by $W = \varepsilon E_b + \rho H$, where H is the incident flux.

For the furnace wall, it is $W_f = \varepsilon_f E_{bf} + \rho_f H_f$. Since the furnace is a black body, it is $\rho_f = 0$ and $\varepsilon_f = 1$. Hence $W_f = E_{bf}$.

For the pipes, the assumption of grey surface means that all the radiation absorbed is re-emitted, $\alpha_p = \varepsilon_p$, and therefore $\rho_p = 1 - \alpha_p = 1 - \varepsilon_p = 1 - 0.5 = 0.5$. The radiative heat flux leaving the pipes is:

$$W_p = \varepsilon_p E_{bp} + \rho_p H_p = \varepsilon_p E_{bp} + (1 - \varepsilon_p) H_p = \varepsilon_p E_{bp} + (1 - \varepsilon_p)(W_p F_{pp} + W_f F_{pf})$$

hence

$$(1 - (F_{pp} - \varepsilon_p F_{pp})) W_p = \varepsilon_p E_{bp} + (1 - \varepsilon_p) W_f F_{pf},$$

$$W_p = \frac{\varepsilon_p E_{bp} + (1 - \varepsilon_p) W_f F_{pf}}{1 - (1 - \varepsilon_p) F_{pp}}$$

Substitute $\varepsilon_p = 0.5$ and rearrange:

$$W_p = \frac{0.5 E_{bp} + 0.5 W_f F_{pf}}{1 - 0.5 F_{pp}} = \frac{E_{bp} + W_f F_{pf}}{2 - F_{pp}} = \frac{E_{bp} + E_{bf} F_{pf}}{2 - F_{pp}}$$

Apply summation rule $F_{pf} + F_{pp} = 1$ to denominator:

$$2 - F_{pp} = 1 + (1 - F_{pp}) = 1 + F_{pf}$$

Substitute and apply Stefan-Boltzmann law:

$$W_p = \frac{E_{bp} + E_{bf} F_{pf}}{1 + F_{pf}} = \frac{\sigma(T_p^4 + T_f^4 F_{pf})}{1 + F_{pf}}$$

[15 marks]

(b)

For two concentric cylinders the view factors are:

$$F_{pp} = 0 \quad (\text{convex surface})$$

$$F_{pf} = 1 \quad (\text{summation rule})$$

$$F_{fp} = r_1/r_2 = 0.1 \quad (\text{given})$$

$$F_{ff} = 1 - r_1/r_2 = 0.9 \quad (\text{summation rule}).$$

Heat balance on the pipe, including heat flux $\dot{Q}_{out}$, is given by:

$$A_p H_p = A_p W_p + \dot{Q}_{out}.$$

Divide both sides by A_p and rearrange:

$$H_p - W_p = \frac{\dot{Q}_{out}}{A_p}$$

Incident flux on the pipe:

$$H_p = W_p F_{pp} + W_f F_{pf} = W_f$$

Substitute:

$$W_f - W_p = \frac{\dot{Q}_{out}}{A_p}$$

Substitute for W_f (black body) and W_p (given in part a) and rearrange:

$$\sigma \frac{F_{pf}}{1 + F_{pf}} T_p^4 = \sigma F_{pf} \left(1 - \frac{F_{pf}}{1 + F_{pf}} \right) T_f^4 - \frac{\dot{Q}_{out}}{A_p}$$

Using $A_p = 2\pi DL$ it is $\frac{\dot{Q}_{out}}{A_p} = \frac{1}{\pi D} \frac{\dot{Q}_{out}}{L}$ with $\frac{\dot{Q}_{out}}{L} = 8 \text{ kWm}^{-1}$:

$$T_p^4 = T_f^4 - \frac{8000}{(5.67 \times 10^{-8})(0.5)(\pi \times 0.2)}$$

$$T_p = 861 \text{ K}$$

[7 marks]

(c) A “black body” from the point of view of radiative heat transfer, is not black in the optical sense, as the emission and absorption properties must be evaluated over the entire spectrum of the electromagnetic radiation, which includes the infrared range. For instance, snow and paper look white in the visible range, but they behave as black bodies in the infrared range.

[3 marks]

QUESTION 4

(a) Two different forms of condensation are observed: film condensation and dropwise condensation. In film condensation, the condensed vapour forms a continuous liquid film on the surface. In dropwise condensation, the condensed vapour forms isolated droplets on the surface. In film condensation, the surface is covered by the liquid film which causes resistance to heat transfer. In dropwise condensation, the droplets slide down due to gravity when they reach a critical size, leaving the surface exposed for additional vapour condensation. As a result, heat transfer rates are larger with dropwise condensation.

[3 marks]

(b)

(i) Laminar film condensation is gravity-controlled when the velocity of the vapour is negligible; it is shear-controlled when the velocity of the vapour is sufficiently large that the interfacial shear stress dominates over gravitational effects. In the first case, it is assumed that $\frac{\partial u}{\partial y}\big|_{y=\delta} = 0$. In the second case, it is assumed that $\frac{\partial u}{\partial y}\big|_{y=\delta} = \tau_i$ and that the terms $\rho_l g$ and

$\frac{\partial p}{\partial x} \approx \rho_v g$ can be neglected from the momentum equation in the x-direction.

(ii) Apply conservation of energy to a differential element of condensate film of length dx : rate of heat transfer from the vapour into the liquid film, dQ , is given by rate of energy release due to condensation at the liquid-vapour interface:

$$dQ = h_{lv} d\dot{m}$$

The rate of heat transfer across the liquid-vapour interface must equal the rate of heat transfer to the solid surface. Hence

$$dQ = q(w \cdot dx)$$

which gives

$$q = \frac{dQ}{w dx} = \frac{h_{lv} d\dot{m}}{w dx} = h_{lv} \frac{d\Gamma}{dx}$$

Since the liquid temperature distribution is linear (assumption of negligible advection in the film), from Fourier's law we have

$$q = k_l \frac{T_{sat} - T_w}{\delta}$$

Rearrange to get

$$\frac{d\Gamma}{dx} = \frac{k_l (T_{sat} - T_w)}{\delta h_{lv}}$$

Using the expression given for the mass flow rate per unit width: $\Gamma(\delta) = \frac{\rho_l(\rho_l - \rho_v)g\delta^3}{3\mu_l}$, we can obtain

$$\frac{d\Gamma}{dx} = \frac{\rho_l(\rho_l - \rho_v)g\delta^2}{\mu_l} \frac{d\delta}{dx}$$

Combining the two equations for $d\Gamma/dx$, we have

$$\delta^3 d\delta = \frac{k_l \mu_l (T_{sat} - T_w)}{\rho_l(\rho_l - \rho_v)g h_{lv}} dx$$

Integrating once and using boundary condition $\delta(x = 0) = 0$ to determine constant of integration, the final result is obtained:

$$\delta(x) = \left[\frac{4k_l \mu_l (T_{sat} - T_w)x}{\rho_l(\rho_l - \rho_v)g h_{lv}} \right]^{1/4}$$

[15 marks]

(c) The presence of a small quantity of non-condensable gas can create a significant resistance to heat transfer near the vapour/liquid interface. A non-condensable gas reduces the partial pressure of the condensable vapour, because the sum of the partial pressures of steam and the non-condensable gas must equal the (constant) total pressure. Because the interfacial temperature (T_i) is usually the saturation temperature corresponding to the partial

pressure of the condensing vapour at the interface, a reduction in the partial pressure of vapour results in a decrease in T_{sat} and T_i . As a result, the driving force for heat transfer is reduced. The presence of an incondensable gas can give rise to very low condensation heat transfer coefficients.

In order to calculate the total heat flux, one needs to find the temperature at the liquid-vapour interface and the modified heat transfer coefficient. The procedure is (assuming that the rate of condensation is known):

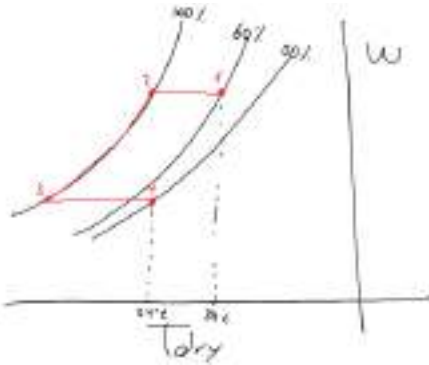
- Find the mole fraction of the condensable component at the interface using mass transfer equation.
- Calculate the partial pressure of the condensable component at the interface and use a relevant physical property table to find the corresponding saturation temperature, which is the interfacial temperature, T_i .
- Calculate the modified heat transfer coefficient using Ackermann correction and appropriate heat transfer correlations in the form $Nu = Nu(Re, Pr)$.

[7 marks]

QUESTION 2

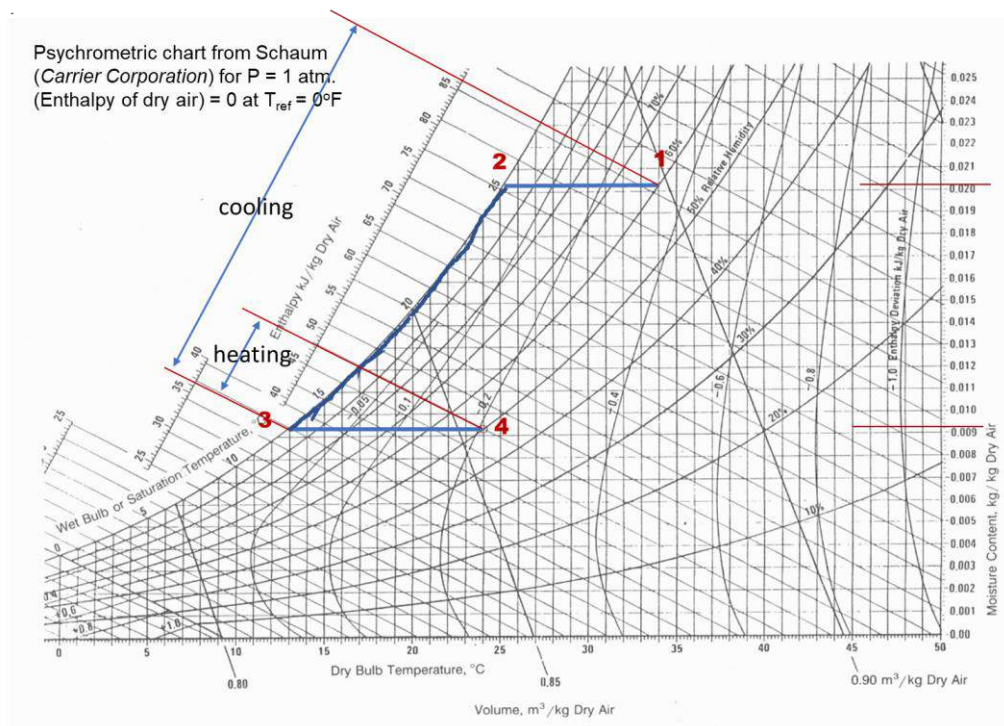
(a)

(i) The air-conditioning process will involve the next steps from 1 to 4:



[5 marks]

(ii) Using the psychrometric chart:



The condensate is $\omega_1 - \omega_4 = 0.0202 - 0.0092 = 0.0110$ kg water (kg dry air) $^{-1}$

Since we have 100 kg dry air h^{-1} , then the condensate per hour is

$$0.0110 \times 100 = 1.10 \text{ kg water h}^{-1}$$

[5 marks]

(iii) Assuming relatively negligible $m_w h_w$ (note $m_w \ll m_a$), the cooling coil has a duty of

$$86.5 - 36.5 = 50 \text{ kJ (kg dry air)}^{-1}$$

Since we have 100 kg dry air h^{-1} , then the heat rate in the cooling coil is

$$50 \times 100 = 5000 \text{ kJ h}^{-1}$$

[5 marks]

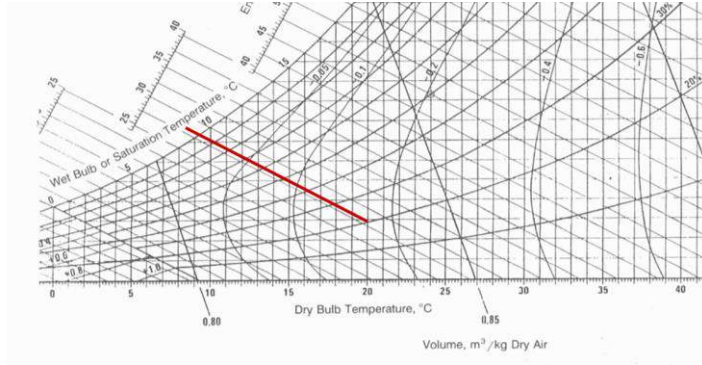
(iv) The heating coil has a duty of $48-36.5=11.5 \text{ kJ (kg dry air)}^{-1}$

Since we have $100 \text{ kg dry air h}^{-1}$, then the heat rate in the heating coil is

$$11.5 \times 100 = 1150 \text{ kJ h}^{-1}$$

[5 marks]

(b) The coldest temperature possible would be the wet bulb temperature of the air used, which is approx. 9.3°C



[5 marks]

QUESTION 3

(a)

$$(i) \quad h'_{lv} = h_{lv} \left(1 + 0.68 \frac{c_{pl}(T_{sat} - T_w)}{h_{lv}} \right) = 2342 \left(1 + 0.68 \left(\frac{(4.182)(340-310)}{2342} \right) \right) =$$

$$= 2342(1 + 0.68(0.05357)) = 2427 \text{ kJ kg}^{-1}$$

$$\bar{\alpha}_N = 0.729 \left[\frac{\rho_l(\rho_l - \rho_v) g k_l^3 h'_{lv}}{N \mu_l (T_{sat} - T_w) D_o} \right]^{\frac{1}{4}} =$$

$$= 0.729 \left[\frac{(987.2)(987.2 - 0.174)(9.81)(0.640)^3 (2427 \times 10^3)}{(10)(577 \times 10^{-6})(340 - 310)(0.050)} \right]^{\frac{1}{4}} =$$

$$= 3753 \text{ W m}^{-2} \text{ K}^{-1}$$

$$(Area) = (10 \text{ tubes} \times \pi \times 0.050 \text{ m} \times 2 \text{ m})$$

$$\dot{Q} = \bar{\alpha}_N (Area) (T_{sat} - T_w) = 3753(10 \times \pi \times 0.050 \times 2)(340 - 310)$$

$$= 353701 \text{ W} = 353.7 \text{ kW}$$

[10 marks]

$$(ii) \quad \dot{m}_{total} = \frac{\dot{Q}}{h_{lv}} = \frac{\bar{\alpha}(Area)(T_{sat} - T_w)}{h_{lv}} = \frac{3753(10 \times \pi \times 0.050 \times 2)(340 - 310)}{2427 \times 10^3} = \frac{353701}{2427 \times 10^3} =$$

$$= 0.14574 \text{ kg s}^{-1} = 525 \text{ kg h}^{-1}$$

[5 marks]

(iii) At the bottom of the tier, $Re_{\delta} = \frac{2\Gamma}{\mu_l} = \frac{2\dot{m}_{total}}{\mu_l w} = \frac{2(0.14574)}{(577 \times 10^{-6})(2)} = 253 < 1800$

Therefore, we can assume laminar film condensation controlled by gravity where Nusselt theory can be applied

We also assume negligible non-condensable gases in the steam; for example, negligible air presence.

[5 marks]

(b) From Fourier's law, the heat flux at the bottom of the plate is

$$q = \frac{k_l}{\delta}(T_{sat} - T_w) = \frac{0.64}{0.05 \times 10^{-3}} 2 = 25600 \text{ W m}^{-2} = 25.6 \text{ kW m}^{-2}$$

[5 marks]

QUESTION 4

(a)

The view factors between concentric spheres are

By observation:

$$F_{22}=0; F_{21}=1$$

To find out F_{12} we can use reciprocity rule:

$$A_1 F_{12} = A_2 F_{21}$$

$$F_{12} = A_2 F_{21} / A_1 = A_2 / A_1 = R_2^2 / R_1^2$$

Since $R_1 = 2R_2$

$$F_{12} = R_2^2 / (2R_2)^2 = 1/4 = 0.25$$

And by summation rule:

$$F_{12} + F_{11} = 1;$$

$$\text{Then } F_{11} = 1 - F_{12} = 1 - R_2^2 / R_1^2 = 0.75$$

To sum up: $F_{11} = 0.75$; $F_{12} = 0.25$; $F_{22}=0$; $F_{21}=1$

[5 marks]

(b) The net radiation between surfaces:

$$\dot{Q}_{net12} = A_1 F_{12} \sigma (T_1^4 - T_2^4) = 4\pi R_1^2 0.25 \sigma (T_1^4 - T_2^4) = \pi R_1^2 \sigma (T_1^4 - T_2^4)$$

Or

$$\dot{Q}_{net12} = A_1 F_{12} \sigma (T_1^4 - T_2^4) = A_2 F_{21} \sigma (T_1^4 - T_2^4) = 4\pi R_2^2 \sigma (T_1^4 - T_2^4)$$

[5 marks]

(c)

$$W_1 = \varepsilon_1 E_{b1} + \rho_1 (W_1 F_{11} + W_2 F_{12})$$

$$W_2 = \varepsilon_2 E_{b2} + \rho_2 (W_1 F_{21} + W_2 F_{22})$$

Substituting F values,

$$W_1 = \varepsilon_1 E_{b1} + \rho_1 (W_1 0.75 + W_2 0.25)$$

$$W_2 = \varepsilon_2 E_{b2} + \rho_2 (W_1)$$

Solving:

$$\begin{aligned} W_1 &= \varepsilon_1 E_{b1} + \rho_1 (W_1 0.75 + W_2 0.25) = \varepsilon_1 E_{b1} + \rho_1 W_1 0.75 + \rho_1 W_2 0.25 = \\ &= \varepsilon_1 E_{b1} + \rho_1 W_1 0.75 + \rho_1 0.25 (\varepsilon_2 E_{b2} + \rho_2 W_1) = \\ &= \varepsilon_1 E_{b1} + \rho_1 W_1 0.75 + \varepsilon_2 E_{b2} \rho_1 0.25 + \rho_2 W_1 \rho_1 0.25 \end{aligned}$$

$$W_1 - \rho_2 W_1 \rho_1 0.25 - \rho_1 W_1 0.75 = \varepsilon_1 E_{b1} + \varepsilon_2 E_{b2} \rho_1 0.25$$

$$W_1 = \frac{\varepsilon_1 E_{b1} + \varepsilon_2 E_{b2} \rho_1 0.25}{1 - \rho_2 \rho_1 0.25 - \rho_1 0.75}$$

$$W_2 = \varepsilon_2 E_{b2} + \frac{\rho_2 \varepsilon_1 E_{b1} + \rho_2 \varepsilon_2 E_{b2} \rho_1 0.25}{1 - \rho_2 \rho_1 0.25 - \rho_1 0.75}$$

where $\rho_1 = 1 - \varepsilon_1$; $\rho_2 = 1 - \varepsilon_2$.

Using

$$\varepsilon_1 = \varepsilon_2 = 0.8$$

Stefan-Boltzmann constant is $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

$$T_1 = 300 \text{ K}$$

$$T_2 = 500 \text{ K}$$

And $E_b = \sigma T^4$

$$W_1 = \frac{\varepsilon_1 E_{b1} + \varepsilon_2 E_{b2} \rho_1 0.25}{1 - \rho_2 \rho_1 0.25 - \rho_1 0.75} = \frac{0.8(5.67 \times 10^{-8} \times 300^4) + 0.8(5.67 \times 10^{-8} \times 500^4) 0.2 \times 0.25}{1 - 0.2 \times 0.2 \times 0.25 - 0.2 \times 0.75}$$

$$W_1 = 606.15 = 606 \text{ Wm}^{-2}$$

$$W_2 = \varepsilon_2 E_{b2} + \rho_2 (W_1) =$$

$$W_2 = 0.8(5.67 \times 10^{-8} \times 500^4) + 0.2 \times 606.15 = 2956.2$$

$$W_2 = 2956 \text{ Wm}^{-2}$$

[15 marks]

SOLUTION KEY – TRANSFER PROCESSES 2 2021

QUESTION 1

(a) The θ -component momentum equation reads as follows:

$$\frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_\theta}{\partial \theta} + u_z \frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} = -\frac{1}{\rho r} \frac{\partial p}{\partial \theta} + \nu \left\{ \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial (r u_\theta)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} + \frac{\partial^2 u_\theta}{\partial z^2} \right\} + g_\theta$$

[1 mark for each assumption – 8 marks in total]

- Steady-state: $\frac{\partial}{\partial t} = 0$ [drop 1st term LHS]
- Flow is unidirectional: $u_\theta \neq 0$, $u_r = u_z = 0$ [drop: 2nd, 4th and 5th term LHS; 4th term RHS]
- Flow is fully developed: $\frac{\partial u_\theta}{\partial \theta} = 0$ [drop 3rd term LHS and 3rd term RHS]
- Pressure depends on z only: $\frac{\partial p}{\partial \theta} = 0$ [drop 1st term RHS]
- Gravity acts vertically downward: $g_\theta = 0$ [drop 6th term RHS].

The z -component momentum equation reads as follows:

$$\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_z}{\partial \theta} + u_z \frac{\partial u_z}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2} \right\} + g_z$$

- Steady-state: $\frac{\partial}{\partial t} = 0$ [drop 1st term LHS]
- Flow is unidirectional: $u_\theta \neq 0$, $u_z = 0$ [drop: 2nd, 3rd, 4th LHS; 2nd 3rd, 4th terms RHS]
- Gravity acts vertically downward: $g_z = -g$.

The simplified equations read as follows.

For the θ -component: **[1 mark]**

$$0 = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial (r u_\theta)}{\partial r} \right) + \frac{\partial^2 u_\theta}{\partial z^2}$$

For the z -component: **[1 mark]**

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g$$

(b) As indicated in the question, $u_\theta = r f(z)$, meaning that:

$$\frac{u_\theta}{r} = f(z)$$

$$\frac{\partial u_\theta}{\partial r} = f(z)$$

Upon expansion of the θ -component equation we obtain: **[2 marks]:**

$$\frac{\partial}{\partial r} \left(\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r} \right) + \frac{\partial^2 u_\theta}{\partial z^2} = 0$$

The first term on the LHS vanishes because the two terms in the parentheses do not depend on r .

One can therefore now solve for $f(z)$ [4 marks]:

$$\frac{d^2 f}{dz^2} = 0$$

subject to the following boundary conditions at the lower and upper plate (note that $u_\theta = \Omega r$ and $f = u_\theta/r$) [2 marks]:

$$\text{At } z = 0, f = 0$$

$$\text{At } z = h, f = \Omega$$

After integrating twice, the following solution is obtained: [2 mark]

$$u_\theta = \Omega r \frac{z}{h}$$

- (c) The pressure distribution is obtained upon integrating the z-component momentum balance. By assuming a reference pressure $p = p_0$ at $z = 0$, we obtain: [2 marks]

$$p = p_0 - \rho gh$$

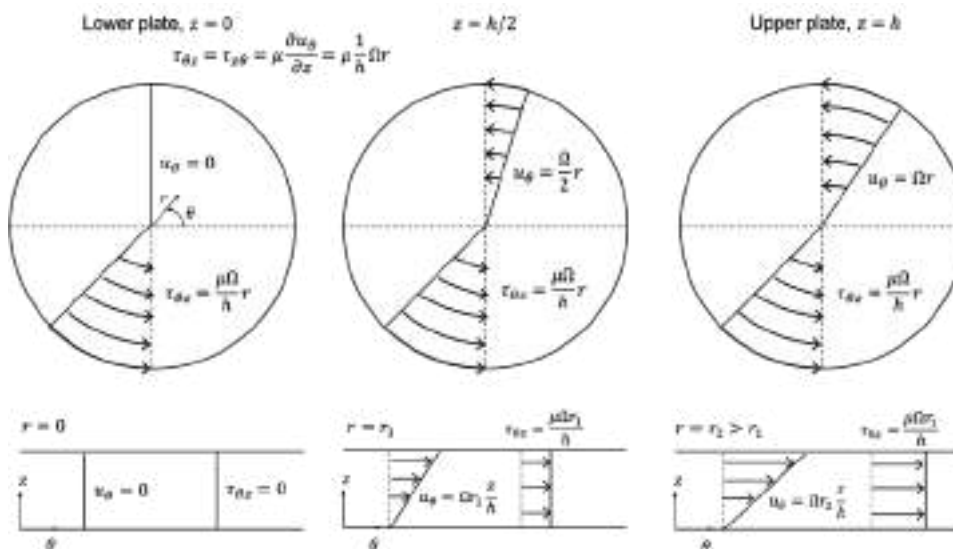
- (d) The relevant stress component is $\tau_{z\theta}$ [2 marks]:

$$\tau_{\theta z} = \tau_{z\theta} = \mu \left(\frac{\partial u_\theta}{\partial z} + \frac{1}{r} \frac{\partial u_z}{\partial \theta} \right)$$

Upon simplifying, we obtain: [2 marks]

$$\tau_{\theta z} = \tau_{z\theta} = \mu \frac{\partial u_\theta}{\partial z} = \mu \frac{1}{h} \Omega r$$

- (e) Velocity and stress profiles [1.5 marks each]:



QUESTION 2

(a)

- (i) Without non-condensable gas we know that the force balance of a bubble is

$$(P_{in} - P_{out})\pi R_b^2 = (2\pi R_b)\sigma ; \text{ where } P_{in} = P_v$$

With non-condensable gas, $P_{in} = P_v + P_{nc}$

Then, the force balance will be updated to:

$$(P_v + P_{nc} - P_{out})\pi R_b^2 = (2\pi R_b)\sigma$$

Simplifying, we get:

$$P_v + P_{nc} - P_{out} = \frac{2\sigma}{R_b}$$

[5 marks]

- (ii) Integrating $\frac{dP}{dT} = \frac{h_{lv}}{T_{sat}(v_v - v_l)}$ we obtain:

$$\Delta T = T_v - T_{sat} = T_l - T_{sat} = \frac{T_{sat}(v_v - v_l)}{h_{lv}}(P_v - P_l)$$

From (i) we know that $P_v - P_{out} = \frac{2\sigma}{R_b} - P_{nc}$ and $v_v \gg v_l$ so we can neglect v_l

Substituting,

$$\Delta T = T_v - T_{sat} = T_l - T_{sat} = \frac{T_{sat}(v_v - v_l)}{h_{lv}}(P_v - P_l) = \frac{T_{sat}(v_v)}{h_{lv}}\left(\frac{2\sigma}{R_b} - P_{nc}\right)$$

(Note this is for equilibrium conditions)

[5 marks]

- (iii) The bubble will grow if superheat increases or, in other words, if the liquid temperature surrounding the bubble increases. This is because under a higher superheat, heat vaporizes more gas, therefore the bubble will grow. It will collapse for the opposite (for example, when the bubble gets in contact with a cold stream).

We can also consider the following. If the liquid temperature surrounding the bubble results in $T_l - T_{sat}$ larger than $T_v - T_{sat}$, the actual bubble radius (r) will be larger than the equilibrium bubble radius (R_b). Since $r > R_b$, it will grow. For the opposite, it will collapse.

The effect of non-condensable gas can be observed in the previous equilibrium equation of 2a(ii). The effect of non-condensable gas is to reduce the amount of superheat needed for bubble growth.

[5 marks]

(b)

Data reading for Question from Table 1:

Liquid Density	ρ_l	1000/1.053=949.7 kg m ⁻³
Vapour Density	ρ_v	1/1.142=0.875 kg m ⁻³
Latent heat of vaporisation	h_{lv}	2225 kJ kg ⁻¹
Surface tension	σ	0.0566 N m ⁻¹

$$q_{max} = 0.131 \rho_v^{\frac{1}{2}} h_{lv} [\sigma g (\rho_l - \rho_v)]^{\frac{1}{4}}$$

$$= 0.131 \times 0.875^{\frac{1}{2}} \times 2225000 (0.0566 \times 9.81 \times (949.7 - 0.875))^{\frac{1}{4}} = 1.307 MW m^{-2}$$

$$\text{Power} = E I; \text{ then } E = \frac{\text{Power}}{I} = \frac{q_{max} \times \text{Area}}{I} = \frac{q_{max} \times \pi D L}{I} = \frac{1.307 \times 10^6 \times \pi \times 0.01 \times 0.20}{200} = 41 V$$

[7 marks]**QUESTION 3**

(a)

- (i) Liquid properties are estimated at the mean condensate film temperature which is (365+345)/2=355 K. Latent heat and vapour properties are estimated at the saturation temperature, i.e. 365 K)

Data for Question 3 from Table A.6:

Saturation temperature	T_{sat}	365	K
Liquid Density	ρ_l	1000/1.030=970.9	kg m ⁻³
Vapour Density	ρ_v	1/2.121=0.452	kg m ⁻³
Latent heat of vaporisation	h_{lv}	2278	kJ kg ⁻¹
Specific heat capacity	c_{pl}	4.199	kJ kg ⁻¹ K ⁻¹
Liquid viscosity	μ_l	343×10^{-6}	N s m ⁻²
Liquid thermal conductivity	k_l	0.671	W m ⁻¹ K ⁻¹
Gravity	g	9.81	m s ⁻²

$$h'_{lv} = h_{lv} \left(1 + 0.68 \frac{c_{pl}(T_{sat} - T_w)}{h_{lv}} \right) = 2278 \left(1 + 0.68 \left(\frac{(4.199)(365 - 345)}{2278} \right) \right) =$$

$$= 2278(1 + 0.68(0.036866)) = 2335.1 kJ kg^{-1} = 2335.1 \times 10^3 J kg^{-1}$$

$$\bar{\alpha}_{vert} = 0.943 \left[\frac{\rho_l (\rho_l - \rho_v) g k_l^3 h'_{lv}}{\mu_l (T_{sat} - T_w) L} \right]^{\frac{1}{4}}$$

$$= 0.943 \left[\frac{970.9(970.9 - 0.452) 9.81 0.671^3 2335.1 \times 10^3}{343 \times 10^{-6} (365 - 345) 0.45} \right]^{\frac{1}{4}} =$$

$$= 6392.9 W m^{-2} K^{-1}$$

$$\dot{Q} = \bar{\alpha} A (T_{sat} - T_w) = 6392.9 W m^{-2} K^{-1} \times (0.45 \times \pi \times 0.060) m^2 \times (365 - 345) K =$$

$$= 10845 W = 10.84 kW$$

We assume **laminar** film condensation controlled by gravity where Nusselt theory can be applied. We also assume negligible **non-condensable** gases in the steam; for example, negligible air presence.

[8 marks]

(ii)

$$\dot{m}_{total} = \frac{\dot{Q}}{h'_{lv}} = \frac{10845 \text{ W}}{2335.1 \times 10^3 \text{ J kg}^{-1}} = 0.004644 \text{ kg s}^{-1} ; \text{ that is, } 16.72 \text{ kg h}^{-1}$$

$$Re_{\delta}|_{x=L} = \frac{4\Gamma|_{x=L}}{\mu_l} = \frac{4\dot{m}_{total}}{\mu_l(\pi D_o)} =$$

$$= \frac{4 \times 0.004644}{343 \times 10^{-6} (\pi \times 0.060)} = 287.3 < 1800 \text{ so laminar Nusselt Theory is valid}$$

[3 marks]

(iii) Observing previous equations, $Re_{\delta} \propto \left(L \times \left(\frac{1}{L}\right)^{\frac{1}{4}}\right)$; that is, $Re_{\delta} \propto L^{3/4}$

$$\text{Then } L \propto Re_{\delta}^{4/3}, \text{ so } L_{trans} = 0.45 \left(\frac{1800}{287.3}\right)^{4/3} = 5.19 \text{ m}$$

[5 marks]

(b)

(i) Comparing

$$\bar{\alpha}_{vert} = 0.943 \left[\frac{\rho_l(\rho_l - \rho_v) g k_l^3 h'_{lv}}{\mu_l(T_{sat} - T_w)L} \right]^{1/4} \quad \text{and} \quad \bar{\alpha}_{N-horiz} = 0.729 \left[\frac{\rho_l(\rho_l - \rho_v) g k_l^3 h'_{lv}}{N \mu_l(T_{sat} - T_w)D_o} \right]^{1/4}$$

We obtain:

$$\bar{\alpha}_{N-horiz} = \frac{0.729}{0.943} \left(\frac{L}{N D_o}\right)^{\frac{1}{4}} \bar{\alpha}_{vert}$$

[3 marks]

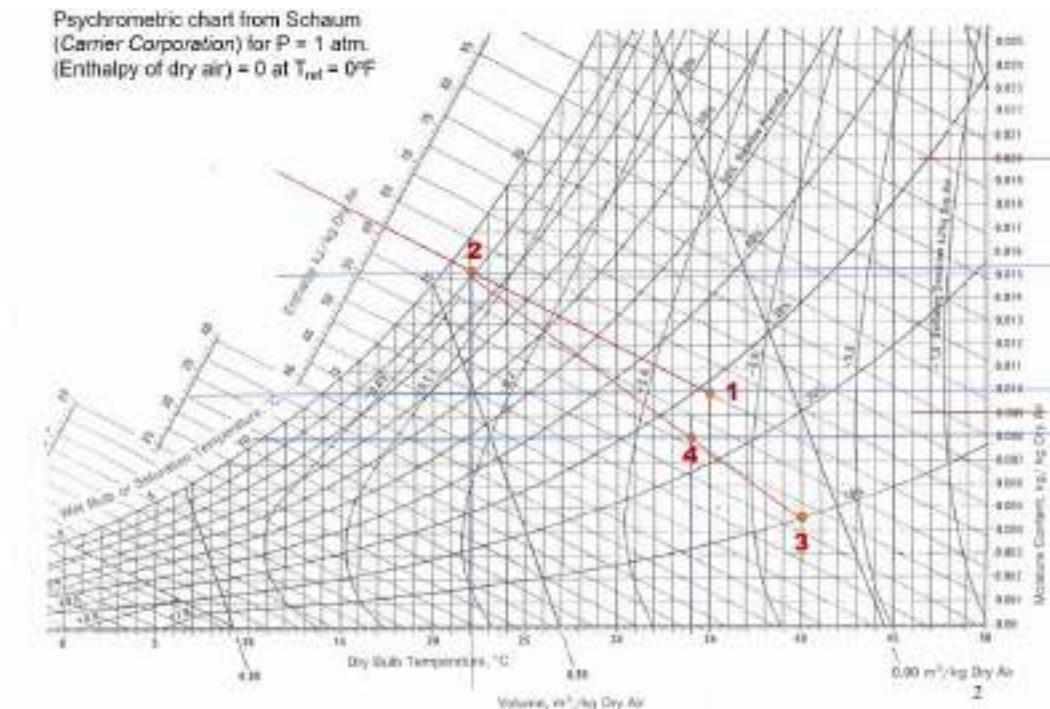
(ii) Applying the previous equation

$$\bar{\alpha}_{N-horiz} = \frac{0.729}{0.943} \left(\frac{0.45}{8 \times 0.060}\right)^{\frac{1}{4}} (6392.9) = 0.761(6392.9) = 4863 \text{ W m}^{-2} \text{ K}^{-1}$$

$$(\dot{m}_{total})_{bank} = 80 \times 0.761(\dot{m}_{total})_{vert} = 80 \times 0.761 \times 0.004644 = 0.2827 \text{ kgs}^{-1} \\ = 1017 \text{ kg h}^{-1}$$

[4 marks]

QUESTION 4



(a)

(i) ω_1 : 0.010 kg water (kg dry air)⁻¹

RH: 28%

h_1 : 61 kJ (kg dry air)⁻¹

[5 marks]

(b) We assume $h_2 = h_1$ since we use a small amount of cold water, and adiabatic

(i) T_{db2} : 22.2°C

T_{wb2} : 21.0°C

[5 marks]

(ii) From graph, ω_2 : 0.0154 kg water (kg dry air)⁻¹

$$m_{a1} + \omega_1 m_{a1} = 10$$

$$\therefore m_{a1} = \frac{10}{(1+\omega_1)} = \frac{10}{(1+0.010)} = 9.9 \text{ kg s}^{-1}$$

The condensate rate is $m_{a1}(\omega_2 - \omega_1) = 9.9 (0.0154 - 0.010) = 0.0535 \text{ kg s}^{-1}$

[5 marks]

(c) Assuming adiabatic

ω_3 : 0.0045 kg water (kg dry air)⁻¹

$$\therefore m_{a3} = \frac{20}{(1+\omega_3)} = \frac{20}{(1+0.0045)} = 19.91 \text{ kg s}^{-1}$$

$$\dot{m}_{a2} + \dot{m}_{a3} = \dot{m}_{a4} = 9.9 + 19.91 = 29.81$$

$$\dot{m}_{a2}\omega_2 + \dot{m}_{a3}\omega_3 = \dot{m}_{a4}\omega_4 = (\dot{m}_{a2} + \dot{m}_{a3})\omega_4$$

$$9.9 \times 0.0154 + 19.91 \times 0.0045 = 29.81\omega_4 \quad ; \quad \omega_4 = 0.0081$$

Reading on graph, $T_{db4} = 34^\circ\text{C}$

h_4 on graph is $55 \text{ kJ (kg dry air)}^{-1}$, which can be confirmed as follows:

$$\dot{m}_{a2}h_2 + \dot{m}_{a3}h_3 = \dot{m}_{a4}h_4 = (\dot{m}_{a2} + \dot{m}_{a3})h_4$$

$$9.9 \times 61 + 19.91 \times 52.5 = 29.81 h_4$$

$$h_4 = 55.3 \text{ kJ (kg dry air)}^{-1}$$

[5 marks]